

UV super-renormalisability of the scalar Brazovskii functional

TECT verification-first repository · claim A3-UV-SUPERRENORMALISABILITY

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1. Purpose and scope

This note establishes the ULTRAVIOLET half of the A3 foundation: the scalar Brazovskii functional in $d = 3$ with the quartic kernel is super-renormalisable (UV-finite). **Scope (narrow, matching A2):** one real scalar field, $d = 3$, $K(q) = \mu^2 + Y(q^2 - q_0^2)^2$, $\mu^2 > 0$; perturbative power counting only. The *stronger* statement that all perturbative correlation functions converge to a cutoff-independent continuum limit is NOT claimed here — it requires a measure definition and a graphwise regulator-convergence lemma and is the separate (open) card A3-PERTURBATIVE-CONTINUUM-CORRELATORS. Numerical checks: [codes/foundations/a3_renormalisation_checks.py](#) (UV asserts).

2. Setup

$F_{\text{TECT}}[\phi] = \int [\frac{1}{2}\phi K(-i\nabla)\phi + \frac{\lambda}{4}\phi^4 + \frac{\gamma}{6}\phi^6]$ with bare propagator

$$G(q) = \frac{1}{K(q)} = \frac{1}{\mu^2 + Y(q^2 - q_0^2)^2} \sim \frac{1}{Y q^4} \quad (q \rightarrow \infty) \quad (1)$$

(a q^{-4} UV decay; assert `propagator_uv_decay_q4`), vertices $\phi^4(\lambda), \phi^6(\gamma)$, momentum cutoff Λ .

3. Power counting

A connected diagram with V vertices, I internal lines, $L = I - V + 1$ loops has, in $d = 3$ with a q^{-4} propagator,

$$D = dL - 4I = 3 - 3V - I \leq -1 < 0 \quad (V \geq 1, I \geq 1) \quad (2)$$

(assert `superficial_degree_all_negative`). The dimension is essential: $D = (d - 4)I - dV + d < 0$ for all $V \geq 1, I \geq 1$ only when $d \leq 3$ ($d = 4$ marginal).

4. Ultraviolet finiteness (Weinberg)

Every loop-containing connected *subdiagram* is itself a connected graph with $V' \geq 1, I' \geq 1$, hence $D' = 3 - 3V' - I' < 0$ by (2). Weinberg's theorem then gives absolute convergence of every loop integral as $\Lambda \rightarrow \infty$ (assert `weinberg_all_subdiagrams_convergent`). Precisely:

No UV-divergent counterterm is required.

A finite normal ordering of the tadpole is an optional scheme choice.

That is, the theory is UV-finite; the counterterm *set* contains no divergent element, and any finite tadpole subtraction is a renormalisation-scheme convention, not a necessity.

5. The tadpole: an illustration (not the full continuum theorem)

The simplest correlator, the equal-point contraction, is the single convergent integral

$$\langle \phi^2 \rangle = \frac{1}{2\pi^2} \int_0^\infty \frac{q^2 dq}{\mu^2 + Y(q^2 - q_0^2)^2} = 0.767, \quad (3)$$

finite because the integrand decays as $1/(Yq^2)$ (assert `tadpole_integral_converges`); its isotropic Riemann sum converges as the spacing $a \rightarrow 0$ (assert `discretisation_to_continuum_convergence`, error $\rightarrow 10^{-12}$). This *illustrates* UV-finiteness for one graph. It does NOT establish lattice-to-continuum convergence of all Feynman graphs, which requires the graphwise dominated-convergence lemma of A3-PERTURBATIVE-CONTINUUM-CORRELATORS.

6. Infrared finiteness

Under $\mu^2 > 0$, $K(q) \geq \mu^2 > 0$, so $G \leq 1/\mu^2$ is bounded: no IR divergence (assert `no_ir_divergence_mu2_positive`). The shell $|q| = q_0$ is a finite enhancement.

Theorem (A3-UV). *In $d = 3$ with $K(q) = \mu^2 + Y(q^2 - q_0^2)^2$, $\mu^2 > 0$, every connected diagram with $V \geq 1, I \geq 1$ has $D = 3 - 3V - I < 0$; by Weinberg's theorem the perturbative expansion is ultraviolet- and infrared-finite. No UV-divergent counterterm is required; a finite normal ordering is an optional scheme choice.*

7. Devil's-advocate (three objections)

- (α) “*Superficial power counting misses nested subdivergences.*” DISMISSED: every subdiagram has $V' \geq 1, I' \geq 1$ hence $D' < 0$; Weinberg requires exactly this and yields absolute convergence.
- (β) “*UV-finiteness implies the continuum limit of all correlators.*” VALID and the reason for the split: it does not, by itself — that needs a measure and a graphwise convergence lemma (separate open card). This note claims only UV-finiteness.
- (γ) “*The finite tadpole is a counterterm.*” DISMISSED: a finite subtraction is a scheme choice, not a divergent counterterm; the precise statement is in §4.

Quantitative sanity check (mandatory): $D = 3 - 3V - I \in \{-1, -2, -5, -6, -7, -10\}$ for the tabulated diagrams; $K(q)/(Yq^4) \rightarrow 1$; $\min_q K = \mu^2 = 5 \times 10^{-3} > 0$; illustrative tadpole 0.767 with refinement error $\rightarrow 10^{-12}$ (6/6 checks).

8. Result footer

Result ID:	A3-UV-SUPERRENORMALISABILITY (scalar, d=3, mu^2>0)
Precise statement:	Every connected diagram with V>=1,I>=1 has D=3-3V-I<0; by Weinberg the perturbative expansion is UV- and IR-finite. No UV-divergent counterterm is required; finite normal ordering is an optional scheme.
Scope:	ONE scalar field, d=3, quartic kernel, mu^2>0; PERTURBATIVE power counting. UV-finiteness ONLY.
Dependencies:	A1-KERNEL-CONV.
Hypotheses:	(H1) d=3 with q^4 kernel [D<0]; (H2) mu^2>0 [IR].
Evidence grade:	PROVED CONDITIONAL (analytic) + EXECUTED (UV checks).
Reproduction command:	python codes/foundations/a3_renormalisation_checks.py
Expected output:	6/6 PASS; exit 0.

Falsification gate: A connected diagram with $V \geq 1, I \geq 1$ and $D \geq 0$ (needs $d \geq 4$
 or a softer kernel); or a UV/IR divergence at $\mu^2 > 0$.
 Tier before / after: A3-UV T1 / T6 PROVED CONDITIONAL -- submitted.
 No-overclaim statement: NOT the continuum limit of correlators (separate card
 A3-PERTURBATIVE-CONTINUUM-CORRELATORS); NOT constructive;
 NOT full Class-II; NOT $\mu^2 < 0$.
 Next required action: Operator sign-off; then the graphwise convergence lemma
 (A3-PERTURBATIVE-CONTINUUM-CORRELATORS).